# 1 Logic

Partially based on WILLIAM M. FARMER & JOSHUA D. GUTTMAN's article
"A Set Theory with Support for Partial Functions".
For simplicity of understanding by the reader, the notation from the article is used here.

**Definition 1.** Value of a function.

$$f(x) \simeq \big(\iota y : \langle x, y\rangle \in f\big)$$

**Theorem 1.** If $f$ is univocal, then

$$\big(f(x) \simeq y\big) \Leftrightarrow \big(\langle x, y\rangle \in f\big)$$

**Proof.**
($\Leftarrow$) (Obvious part) Here we use axiom PFOL10′ which is equaivalent to PFOL10:

$$(\iota x : \varphi)\downarrow \quad \overset{\text{PFOL10}'}{\Longleftrightarrow} \quad (\exists! x : \varphi)$$

So the proof is the following:

$$\big(\langle x, y\rangle \in f\big) \quad \overset{\exists\text{-intro}}{\Longrightarrow} \quad \big(\exists w : \langle x, w\rangle \in f\big)$$

$$f \text{ univocal} \quad \overset{\text{def}}{\Longleftrightarrow} \quad \big(!w : \langle x, w\rangle \in f\big)$$

$$\big(\exists w : \langle x, w\rangle \in f\big) \quad \& \quad \big(!w : \langle x, w\rangle \in f\big) \quad \overset{\exists!\text{-intro}}{\Longrightarrow} \quad \big(\exists! w : \langle x, w\rangle \in f\big)$$

$$\big(\exists! w : \langle x, w\rangle \in f\big) \quad \overset{\text{PFOL10}'}{\Longrightarrow} \quad \big(\iota w : \langle x, w\rangle \in f\big)\downarrow$$

$$\big(\iota w : \langle x, w\rangle \in f\big)\downarrow \quad \overset{\text{PFOL11}}{\Longrightarrow} \quad \langle x, (\iota w : \langle x, w\rangle \in f)\rangle \in f$$

$$\langle x, (\iota w : \langle x, w\rangle \in f)\rangle \in f \quad \& \quad \big(\langle x, y\rangle \in f\big) \quad \overset{f \text{ univocal}}{\Longrightarrow} \quad (\iota w : \langle x, w\rangle \in f) \simeq y$$

$$f(x) \simeq y$$

($\Rightarrow$)

$$(y\downarrow) \quad \text{(by PFOL8)}$$

$$(y\downarrow) \quad \& \quad ((\iota w : \langle x, w\rangle \in f) \simeq y) \quad \overset{\text{PFOL7}}{\Longrightarrow} \quad (\iota w : \langle x, w\rangle \in f)\downarrow$$

$$\big(\iota w : \langle x, w\rangle \in f\big)\downarrow \quad \overset{\text{PFOL11}}{\Longrightarrow} \quad \langle x, (\iota w : \langle x, w\rangle \in f)\rangle \in f$$

$$\langle x, (\iota w : \langle x, w\rangle \in f)\rangle \in f \quad \& \quad \big((\iota w : \langle x, w\rangle \in f) \simeq y\big) \quad \overset{\text{PFOL7}}{\Longrightarrow} \quad \big(\langle x, y\rangle \in f\big)$$

**Done.**